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Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 27-03-2026

Automated Quiz Engine with PDF Certification Generation

SDG: SDG No – SDG Title(Example: SDG 4 – Quality Education)

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Courses Handling Faculty:

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INTRODUCTION



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- Many online quiz systems lack automation and proper result management, making evaluation slow and inefficient.
- Manual quiz correction and certificate generation require time and human effort.
- This project solves these problems by automating quiz conduction, scoring, and certification.
- It is important because it reduces human errors and saves time in educational and training environments.
- The system has real-world relevance in schools, colleges, and online learning platforms.
- Users can register, login, and attempt quizzes dynamically from a database.
- The system automatically calculates scores and stores results securely.
- It also generates a PDF certificate for users who pass the quiz, making it a complete automated solution

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PROBLEM STATEMENT



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The current quiz systems in many educational and training environments are not fully automated and often rely on manual processes for evaluation and certification. This leads to delays, errors in scoring, and inefficient management of user data and results. Additionally, there is no integrated mechanism to instantly generate certificates after successful completion of quizzes. This creates inconvenience for both administrators and users. Therefore, there is a need for a simple, reliable, and automated quiz system that can conduct exams, evaluate responses, store results, and generate certificates efficiently to improve accuracy and performance.

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OBJECTIVE(S)



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To design and develop an automated quiz system for conducting tests and evaluating user responses.

To implement automatic result calculation and PDF certificate generation based on user performance.

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PROPOSED SYSTEM



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- The system is a web-based automated quiz engine that allows users to register, login, and attempt quizzes online.

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- Questions are stored in a MySQL database and fetched dynamically, ensuring flexibility and easy updates.

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- The system automatically evaluates user responses and calculates scores instantly.

- Results are stored securely in the database for future reference and analysis.

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- A PDF certificate is generated automatically for users who meet the passing criteria.

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- It eliminates manual evaluation, reducing errors and saving time.

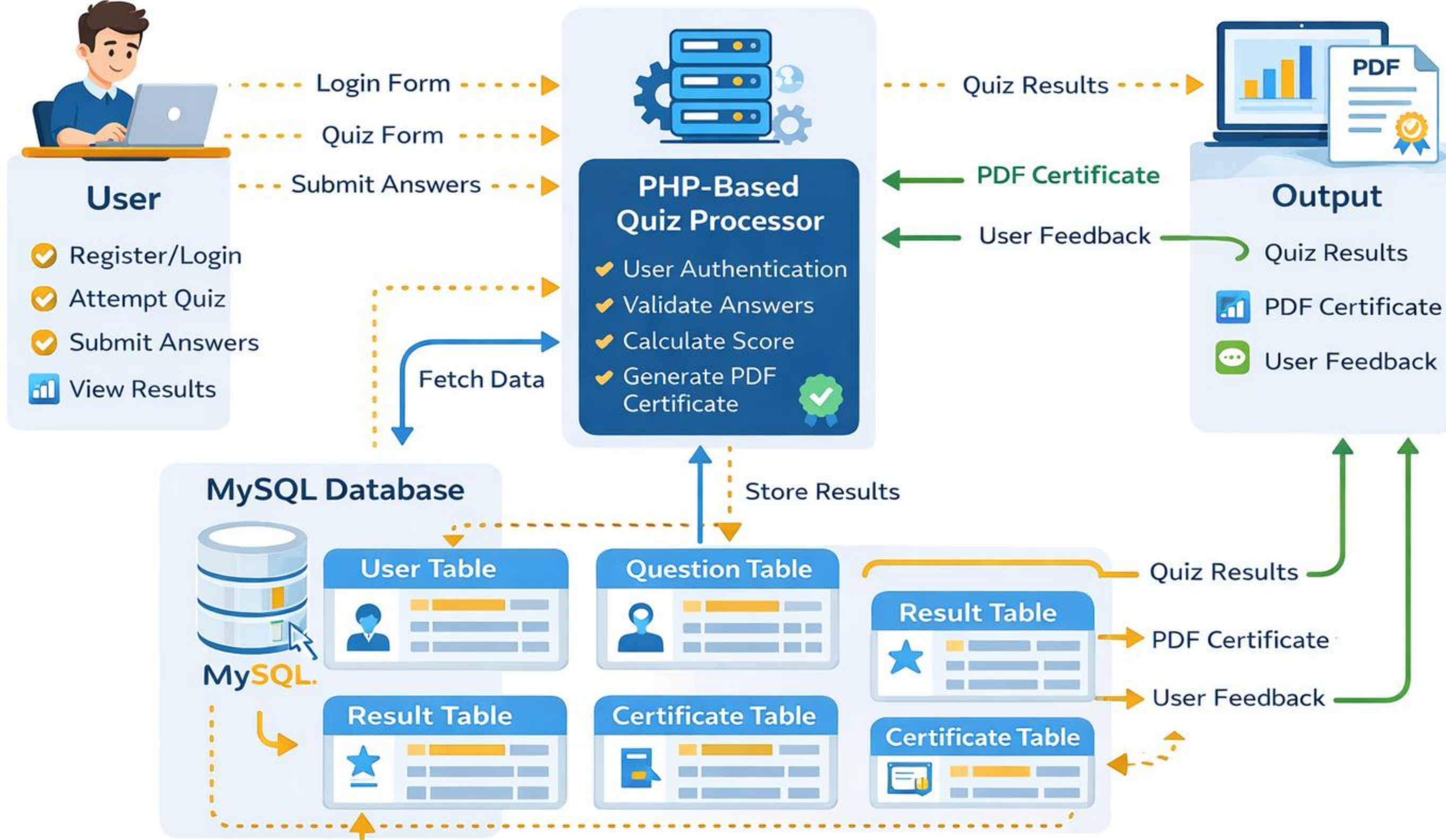
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- The system provides faster processing and improves efficiency compared to traditional methods.

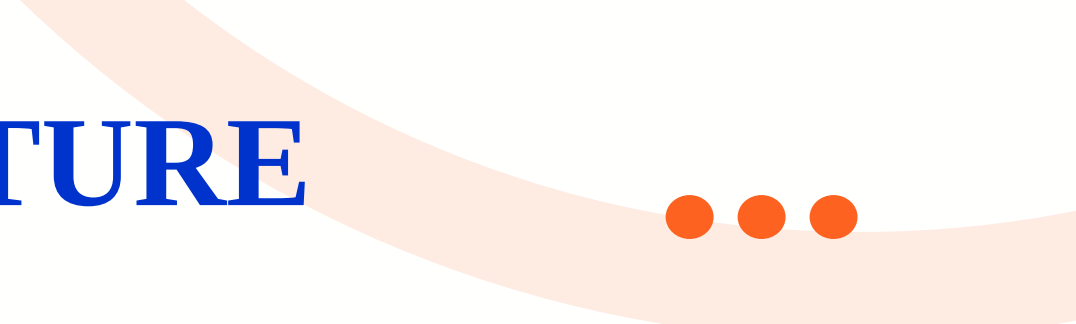
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System Architecture



PROPOSED SYSTEM ARCHITECTURE



1. Input

User registers and logs in

User selects and answers quiz questions

Data entered through HTML forms

2. Processing

PHP handles user authentication

Quiz answers are validated

Score is calculated automatically

Pass/fail condition is checked

3. Database

MySQL stores:

User details

Questions and answers

Quiz results

Data is retrieved and updated dynamically

4. Output

Displays quiz score instantly

Shows result (Pass/Fail)

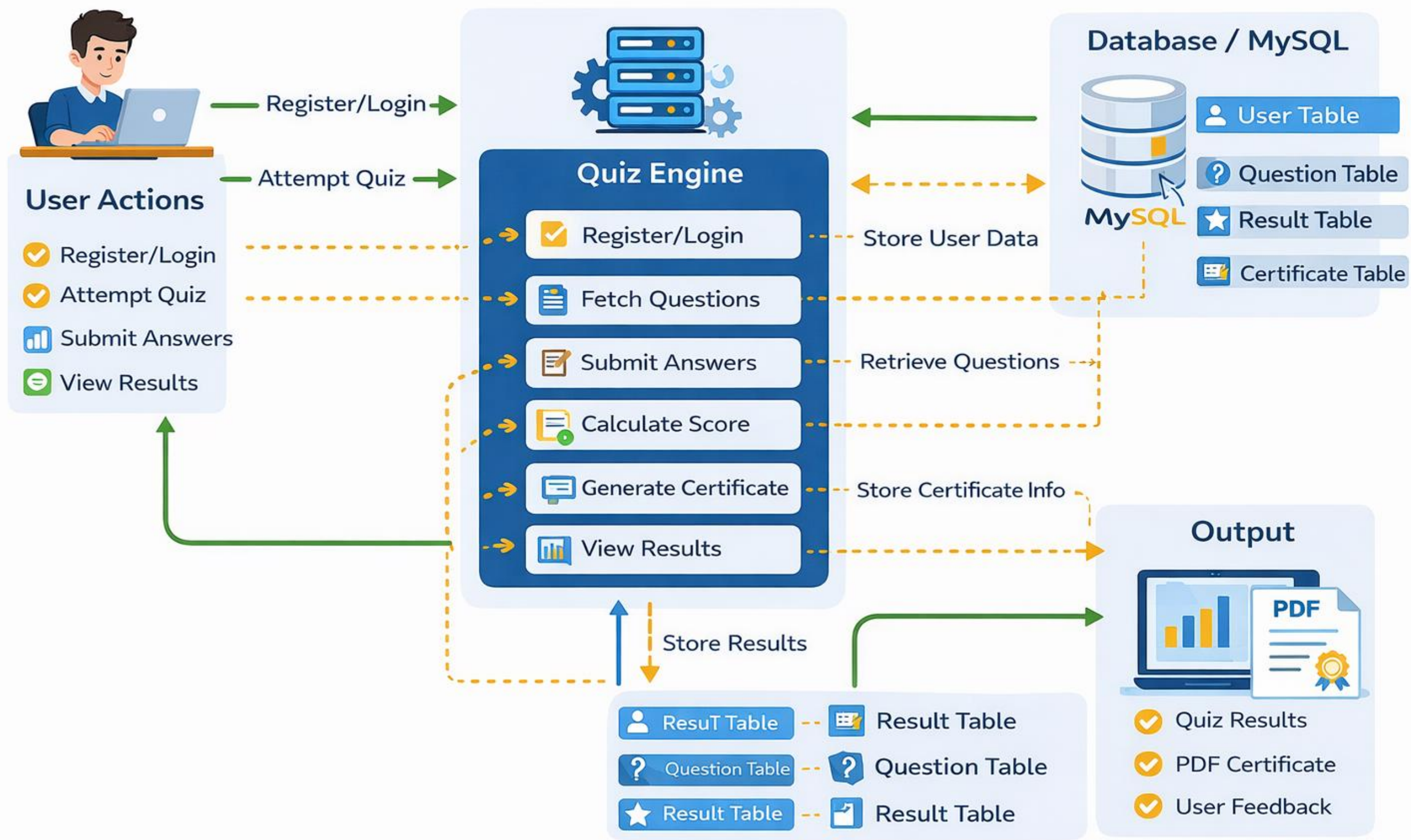
Generates downloadable PDF certificate

Provides user feedback

WORKFLOW



Data Flow Diagram



TECHNOLOGIES USED



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Frontend: HTML5 , CSS3 , JavaScript

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Backend: PHP

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Database: MySQL

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Tools: XAMPP Server , phpMyAdmin , Visual Studio Code (or any code editor) ,Web Browser (Chrome/Edge)

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Hardware/Software Requirements:

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Hardware: Minimum 4 GB RAM , Intel i3 or above processor ,10 GB free disk space

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Software: Windows 10/11 , XAMPP (Apache + MySQL + PHP) , Web Browser (Chrome/Edge)

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Code Editor (VS Code / Notepad++)

MODULE DESCRIPTION



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Module 1: User Management

Handles user registration, login, and logout functionality. It manages user credentials and maintains session control for secure access to the quiz system.

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Module 2: Quiz Processing

Implements the core quiz functionality by displaying questions, capturing user responses, and evaluating answers. It calculates scores and determines pass/fail status.

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Module 3: Database Management

Manages storage and retrieval of data such as user details, quiz questions, and results using MySQL. Ensures data consistency and efficient querying.

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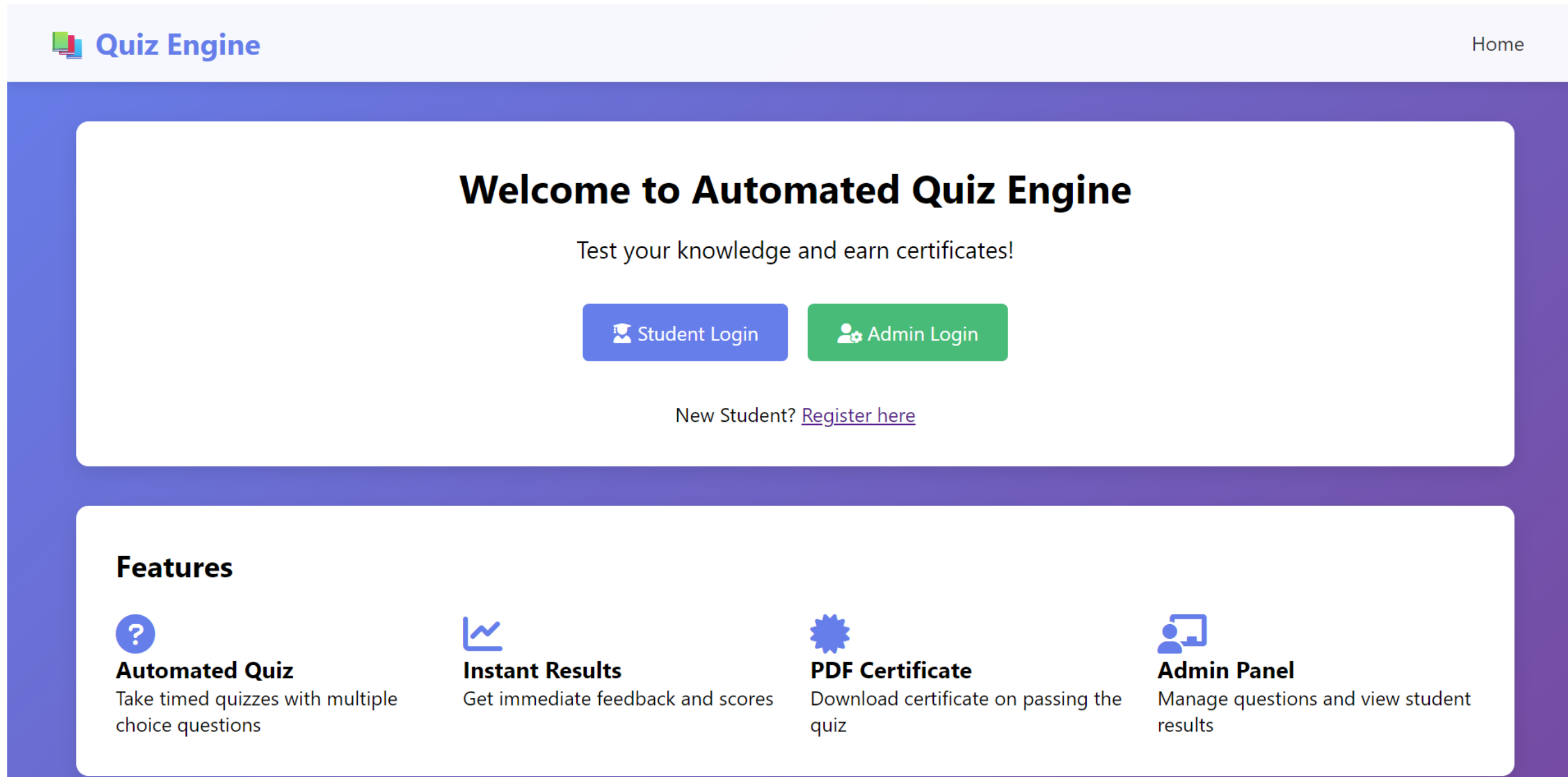
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Module 4: Result & Certification

Displays quiz results instantly and generates a PDF certificate for users who pass the quiz. Stores result history for future reference.

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IMPLEMENTATION SCREENSHOTS



User-friendly Automated Quiz Engine interface with student and admin login options. Provides instant results, quiz access, and PDF certificate generation features

IMPLEMENTATION SCREENSHOTS



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Quiz Engine

Home Student Login Register

Student Login

Username

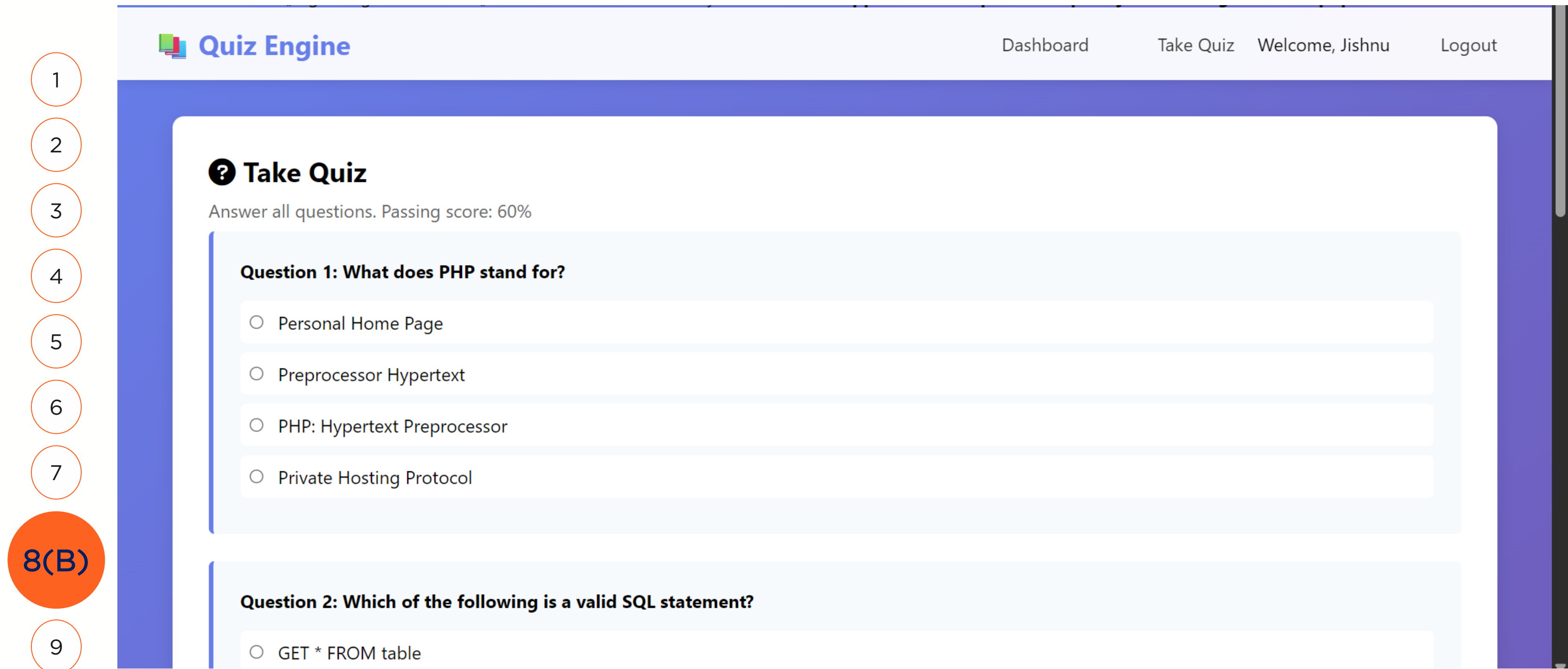
Password

Login

Don't have an account? [Register here](#)

Secure student login interface for accessing the quiz system.
Allows registered users to authenticate and start quizzes seamlessly

IMPLEMENTATION SCREENSHOTS



Interactive quiz interface displaying multiple-choice questions to the user.
Evaluates user responses in real-time to calculate score and determine pass status.

IMPLEMENTATION SCREENSHOTS



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CERTIFICATE OF ACHIEVEMENT

This certificate is proudly presented to

Jishnu

For successfully completing the quiz with a score of

5/5 (100.00%)

Demonstrating excellent knowledge and understanding of the subject matter.

Quiz System Admin
Authorized Signatory

Date: March 27, 2026

Automatically generated certificate awarded upon successful quiz completion.
Displays user name, score, and date as proof of achievement

ADVANTAGES & LIMITATIONS



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ADVANTAGES

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1. Automates quiz conduction, evaluation, and certificate generation, reducing manual effort.

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2. Provides instant results, saving time for both users and administrators.

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3. Reduces human errors in scoring and result processing.

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4. Easy to use and accessible through a web browser.

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5. Stores user data and results securely for future reference.

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6. Scalable system that can be extended with more features like admin panel or analytics.

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LIMITATIONS

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1. Requires internet connection or local server to access the system.

2. Basic security implementation (can be improved with encryption and hashing).

3. Limited to objective-type questions (MCQs only).

4. No advanced analytics or reporting features in current version.

5. Performance may decrease with very large number of users or questions.

Thank you

